THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO SECONDARY SCHOOL

FORM TWO PRE-NATIONAL EXAMINATION 1
033 BIOLOGY

Time: 2:30 Hours **April 2026**

Instructions

1. This paper consists of three sections A, B, and C with a total of ten questions.
2. Attempt **ALL** questions.
3. All answers must be written in the spaces provided.
4. Write your examination number on every page in this question paper.
5. All writing must be in blue/black ink. Diagrams must be drawn in pencil and labeled in ink.
6. Cell phones and calculators are not allowed in the examination room

QUESTION NUMBER	SCORE	INITIALS OF EXAMINER
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		

SECTION A (15 MARKS)

1. Select the most correct answer in each item and write its letter beside its corresponding item number.

(i) The state that distinguishes a mango tree from a rock is known as…

A. Observation
B. Life
C. Nature
D. Hypothesis

(ii) Which of the following organelles is correctly matched with its specific function in an animal cell?

A. Chloroplast — Photosynthesis
B. Ribosome — Protein synthesis
C. Cell wall — Protection and shape
D. Large central vacuole — Storage of cell sap

(iii) Seypalm keeps chickens and goats on his farm. The chickens reproduce at a faster rate than the goats, therefore he has so many of them. Distinction betwee Seypalm's livestock as belonging to different groups based on their ability to interbreed and produce fertile offspring is derived from:

A. Belonging to the same Kingdom
B. Belonging to the same Phylum
C. Belonging to the same Genus
D. Belonging to the same Species

(iv) In the human digestive system, the wave-like muscular contractions that push food along the esophagus and intestines are called:

A. Absorption
B. Assimilation
C. Peristalsis
D. Digestion

(v) Which blood vessel carries deoxygenated blood from the heart to the lungs?

A. Pulmonary artery
B. Pulmonary vein
C. Aorta
D. Vena Cava

(vi) During starvation and after when the human body has used up all of the stored fat in the body, where does it get energy source from before dying?

A. Glycogen
B. Protein
C. Carbohydrates
D. Lipids

(vii) The method of preserving food in an edible liquid such as brine and vinegar is called.

A. Curing
B. Fermentation
C. Bio-preservation

D. Pickling

(viii) Which of the following organisms belong to the same classes in the taxonomic hierarchy?

A. bats and birds
B. Pangolins (Scaly anteaters) and monitor lizards
C. Bats and pangolins (Scaly anteaters)
D. Turtles and toads

(ix) Where does gaseous exchange in grasshoppers occur?

A. In spiracles
B. In the lungs
C. In trachea
D. In the gills

(x) Which of the following is present in animal cell but absent in the higher plants cell?

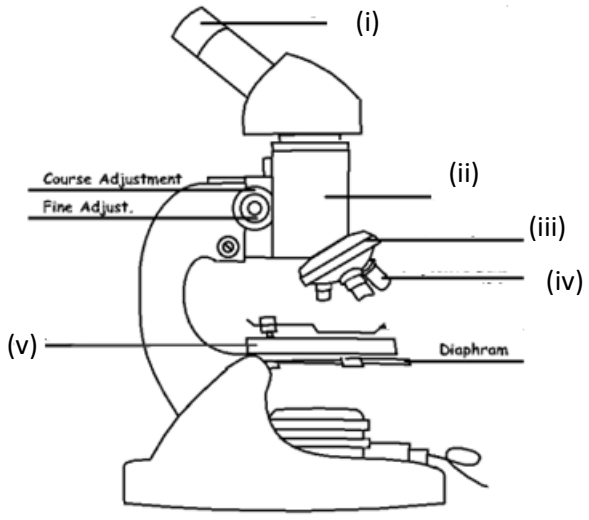
A. Nucleus
B. Ribosome
C. Centriole
D. Mitochondria

2. Match the items in LIST A with the correct responses in LIST B.

LIST A	LIST B
(i) Vitamin A (ii) Vitamin C (iii)Iron (iv)Vitamin D (v) Iodine	A. Bleeding gums and slow healing of wounds. B. Poor night vision or night blindness. C. Soft bones and bowed legs (Rickets). D. Low energy and pale skin (Anaemia). E. Swelling of the neck (Goiter). F. Swollen abdomen and thin limbs (Kwashiorkor). G. Excessive bleeding from a small cut.

SECTION B (70 MARKS)

3. As a biology student, how would you help your friend to identify and state the function of each of the microscope parts labelled A, B, C, D and E?



(i) ……

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(ii) ……

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(iii)……

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(iv)……

(v) ……

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4. (a) Define the term Binomial Nomenclature and state three rules used when writing scientific names of organisms.

(i) Binomial nomenclature:

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(ii) Rules when writing scientific names:

a) ……

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b) ……

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c) ……

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(b) A student found an organism with the following characteristics: It has mammary glands, possesses hair on its body, and gives birth to live young.

(i) To which **Class** does this organism belong?

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(ii) Name the **Kingdom** and **Phylum** of this organism.

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5. (a) Name the four main components of human blood and briefly explain the function of each.

(i) ……

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(ii) ……

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(iii)……

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(iv)……

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(b) Why is the left ventricle of the human heart thicker than the right ventricle?

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6. The rate of gas exchanges in human and other mammals is not always the same between different individuals of the same species, and it is also not constant all the time with the same individual due to certain factors. Briefly explain five of those factors.

(i) ……

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(ii) ……

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(iii)……

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(iv)……

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(v) ……

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7. (a) Give two differences between a prokaryotic cell and eukaryotic cell.

(i) ……

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(ii) ……

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(b) (i) Define cell differentiation and explain its importance

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(ii) Mention two specialized animal cells and two specialized plant cells.

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8. In a meeting in your street, you heard participants and the chairperson calling for a campaign to eliminate all types of bacteria because they are very bad organisms that cause diseases and spoil food. As a biology student, they allowed you to say something about their resolution. What five things can you tell them about bacteria to convince them that it wouldn't be a good idea to eliminate all types of bacteria?

(i) ……

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(ii) ……

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(iii) ……

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(iv) ……

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(v) ……

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9. You are given a food solution sample by your teacher and asked to find out whether the sample contains reducing sugars, non-reducing sugars and starch. Outline laboratory tests you would carry out in order to obey your teacher.

(i) Testing for reducing sugars

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(ii) Testing for non-reducing sugar

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(iii)Testing for starch

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SECTION C (15 MARKS)

10. The human's heart is perfectly adapted to the amazing functions it performs; explain this statement in five points.

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